EXCEL VBA BUSINESS PROCESS AUTOMATION

FULLY AUTOMATED INVENTORY & BILLING ERP SYSTEM

End-to-End Business Process Automation solution in Microsoft Excel (VBA)

Presented by Pradeep Kumar Kurrey

Advanced Excel & MIS Executive



REAL-TIME INVENTORY · GST INVOICING · REPORTING VBA AUTOMATION · 5,000+ AUDIT RECORDS

A Unified Excel Architecture Connecting Stock, Invoicing, and Audits

Eliminates fragmented records by integrating inventory movements, tax billing, and master ledgers into a single , tamper-proof Excel VBA environment.



Centralize:

Single-source master sheets for Products, Customers, and Suppliers.



Automate:

1-Click VBA execution replaces hours of manual calculation.



Validate:

Active data guards prevent out-of-stock billing and entry errors.



Audit:

Automated logging of 5,000+ transactions with historical reprint.

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Manual Spreadsheet Vulnerabilities vs Automated Excel ERP

CHALLENGES

Where work slows down

- Accidental billing without available inventory
- Formula corruption due to manual user tampering
- Slow invoice typing and manual tax calculations
- Missed reorder dates causing stockouts
- Scattered sales records with zero audit trail

OBJECTIVES

What the System Delivers

- Real-time stock deduction on invoice generation
- Protected cells and locked automated formula architecture
- Instant 1-click GST invoice with VBA NumberToWords
- Automated threshold popups with instant supplier details
- Centralized Sales Master with 5,000+ permanent rows

The intended value is a more controlled path from stock movement to billing record.

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End-to-End Relational Data Pipeline Across 8 connected Modules

Each stage adds structure, control, or context before the information reaches a business review.

Master Database:

Centralized Product, Supplier, and Customer records.

Live validation:

Dynamic check intercepts stock deficit before entry.

1-Click Billing:

Auto-fetches prices, computes 18% GST and discounts.

VBA Core Engine:

Executes multi-sheet updates and generates clean PDF/Print.

Ledger Sync:

Automatically updates Opening, Purchase, and Closing Stock.

Master Sales Commit:

Appends transaction row permanently to Sales Master.

Live Implemented Pipeline : 100% verified transactional integrity across all operational sheets.

Live Stock Ledger with Automated Reorder Level Triggers

Live Status Highlight:

Automated conditional formatting marks status as **"Reorder" (Red)** whenever Available Stock falls to or below safety thresholds (e.g. Epson Printer: 5 Pcs, Dell Monitor: 21 Pcs)

Zero Mismatch:

Stock cuts are triggered instantly upon invoice generation.

Ledger Formula:

Closing Stock = Opening Stock + Purchase Stock – Selling Stock

Before Automation

Stock Register

Product ID	Product Name	Category	Brand	Opening Stock	Purchase Stock	Total Stock	Selling Stock	Available Stock	Reorder Level	Status
P0001	Epson Printer	Printer	Epson	18	49	67	52	15	5	OK
P0002	Dell Desktop	Desktop	Dell	15	135	150	47	103	10	OK
P0003	Dell Monitor	Monitor	Dell	27	51	78	47	31	25	OK
P0004	Boat Headset	Headset	Boat	27	59	86	66	20	8	OK
P0005	Sandisk Pen Drive	Pen Drive	SanDisk	19	72	91	38	53	25	OK
P0006	TP-link Bluetooth Adapter	Bluetooth Adapter	TP-link	44	230	274	26	248	20	OK
P0007	Logitech Keyboard	Keyboard	Logitech	27	116	143	58	85	5	OK
P0008	Kingston Ram	RAM	Kingston	18	133	151	39	112	15	OK
P0009	Lg Monitor	Monitor	LG	43	202	245	25	220	20	OK
P0010	Lenovo Laptop	Laptop	Lenovo	41	88	129	45	84	5	OK
P0011	Samsung Monitor	Monitor	Samsung	13	107	95	56	39	8	OK
P0012	Logitech Mouse	Mouse	Logitech	44	188	141	22	102	5	OK
P0013	Samsung Monitor	Monitor	Samsung	16	128	144	42	102	5	OK
P0014	Dell Keyboard	Keyboard	Dell	10	174	184	23	161	5	OK

After VBA Execution (Invoice Generated & Stock Updated

Stock Register

Product ID	Product Name	Category	Brand	Opening Stock	Purchase Stock	Total Stock	Selling Stock	Available Stock	Reorder Level	Status
P0001	Epson Printer	Printer	Epson	18	49	67	62	5	5	Reorder
P0002	Dell Desktop	Desktop	Dell	15	135	150	57	93	10	OK
P0003	Dell Monitor	Monitor	Dell	27	51	78	57	21	25	Reorder
P0004	Boat Headset	Headset	Boat	27	59	86	66	66	8	OK
P0005	Sandisk Pen Drive	Pen Drive	SanDisk	19	72	91	38	53	25	OK
P0006	TP-link Bluetooth Adapter	Bluetooth Adapter	TP-link	44	230	274	26	118	20	OK
P0007	Logitech Keyboard	Keyboard	Logitech	27	116	143	58	85	5	OK
P0008	Kingston Ram	RAM	Kingston	18	133	151	39	112	20	OK
P0009	Lg Monitor	Monitor	LG	43	202	245	25	220	20	OK
P0010	Lenovo Laptop	Laptop	Lenovo	41	88	129	45	84	5	OK
P0011	Samsung Monitor	Monitor	Samsung	44	107	95	56	39	8	OK
P0012	Logitech Mouse	Mouse	Logitech	44	107	151	22	102	5	OK
P0013	Samsung Monitor	Monitor	Samsung	10	128	144	42	102	5	OK
P0014	Dell Keyboard	Keyboard	Dell	10	174	184	23	161	5	OK

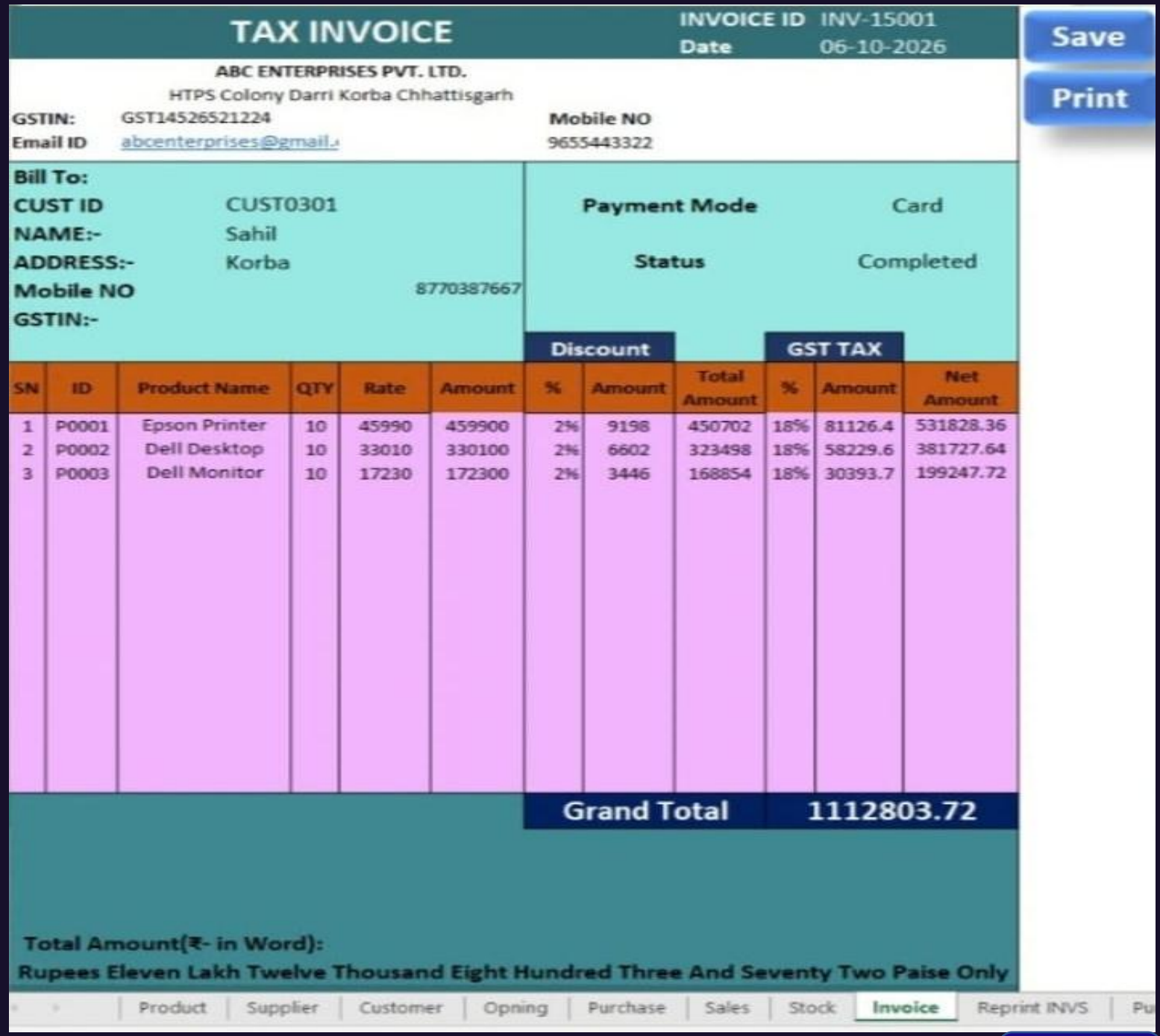
1-Click Tax Invoice Generation with Dynamic Currency Words

CAPTURE: Auto-fetches
Customer (Sahil / CUST0301) and product rates dynamically.

CALCULATE:
Computes line items, discounts (2%), and 18% GST automatically (Grand Total: ₹11,12,803.72).

VBA NUMBER-TO-WORDS:
Built-in custom VBA translates payout into formal words: "Rupees Eleven Lakh Twelve Thousand Eight Hundred Three and Seventy-two Paise Only

1-CLICK ACTION:
Dedicated "Save" and "Print" macro buttons for instant execution.



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How VBA Macros Bridge User Actions with Protected Ledger Updates

The macro is the bridge between an intentional user action and a visible, reviewable outcome.

Event Trigger (Save Button)

User clicks the “Save“ macro button. VBA initializes execution and sets Application. ScreenUpdating=False for instant processing.

Pre-Execution Validation

Macro checks live stock available quantity. If quantity exceeds stock, execution halts immediately with an error prompt

Invoice Engine

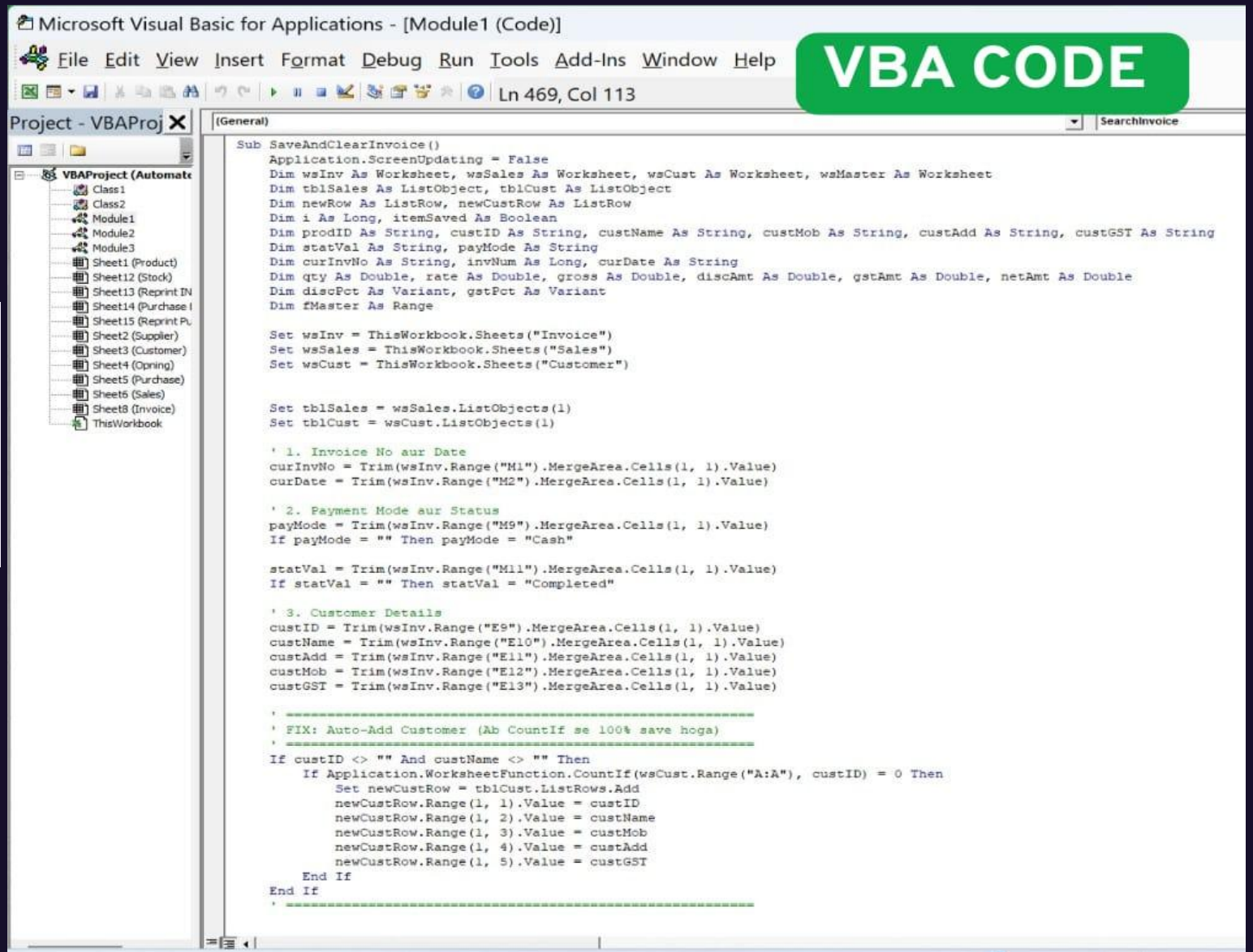
Generates sequential invoice ID (INV-15001), computes tax/ discounts, and converts the grand total into formal Indian currency words.

Multi-Sheet Database

Simultaneously deducts Qty from the Stock Register and appends a permanent transaction row into the Sales Master database.

Auto-Reset & Success State

Clears input fields, restores screen updating and prepares the workbook instantly for the next customer entry



Real-Time Inventory tracking Triggers Automated Reorder Alerts & Vendor Fetch

Total Sales

5,000+ Transactions Logged

Tracked SKUs

15+ Active IT Hardware Products

Stock Accuracy

100% Real-Time Mathematical Balance

Active Reorder Alert

Triggered for Dell Monitor (21 Pcs remaining)

Auto-Fetched Supplier Details

Stock Register

Product ID	Product Name	Category				Total Stock	Selling Stock	Available Stock	Reorder Level	Status
P0001	Epson Printer	Printer				67	62	5	5	Reorder
P0002	Dell Desktop	Desktop				150	57	93	10	OK
P0003	Dell Monitor	Monitor				78	57	21	25	Reorder
P0004	Boat Headset	Headset				86	66	20	8	OK
P0005	Sandisk Pen Drive	Pen Drive				91	38	53	25	OK
P0006	TP-link Bluetooth Adapter	Bluetooth Adapt				274	26	248	20	OK
P0007	Logitech Keyboard	Keyboard				143	58	85	5	OK
P0008	Kingston Ram	RAM				151	39	112	15	OK
P0009	Lg Monitor	Monitor				245	25	220	20	OK
P0010	Lenovo Laptop	Laptop				129	45	84	5	OK
P0011	Samsung Monitor	Monitor				95	56	39	5	OK
P0012	Logitech Mouse	Mouse	Logitech	44	107	151	22	129	8	OK
P0013	Samsung Monitor	Monitor	Samsung	16	128	144	42	102	5	OK
P0014	Dell Keyboard	Keyboard	Dell	10	174	184	23	161	5	OK
P0015	TP-link Wi-Fi Adapter	Wi-Fi Adapter	TP-link	37	33	70	44	26	15	OK

Stock Alert

?? REORDER ALERT!

Product ID: P0003
Product: Dell Monitor

?? STOCK DETAILS
Available Stock: 21 Pcs
Reorder Level: 25 Pcs

?? LATEST SUPPLIER DETAILS
Supplier ID: SUP028
Supplier Name: Shree Suppliers
Contact No: 9689542477

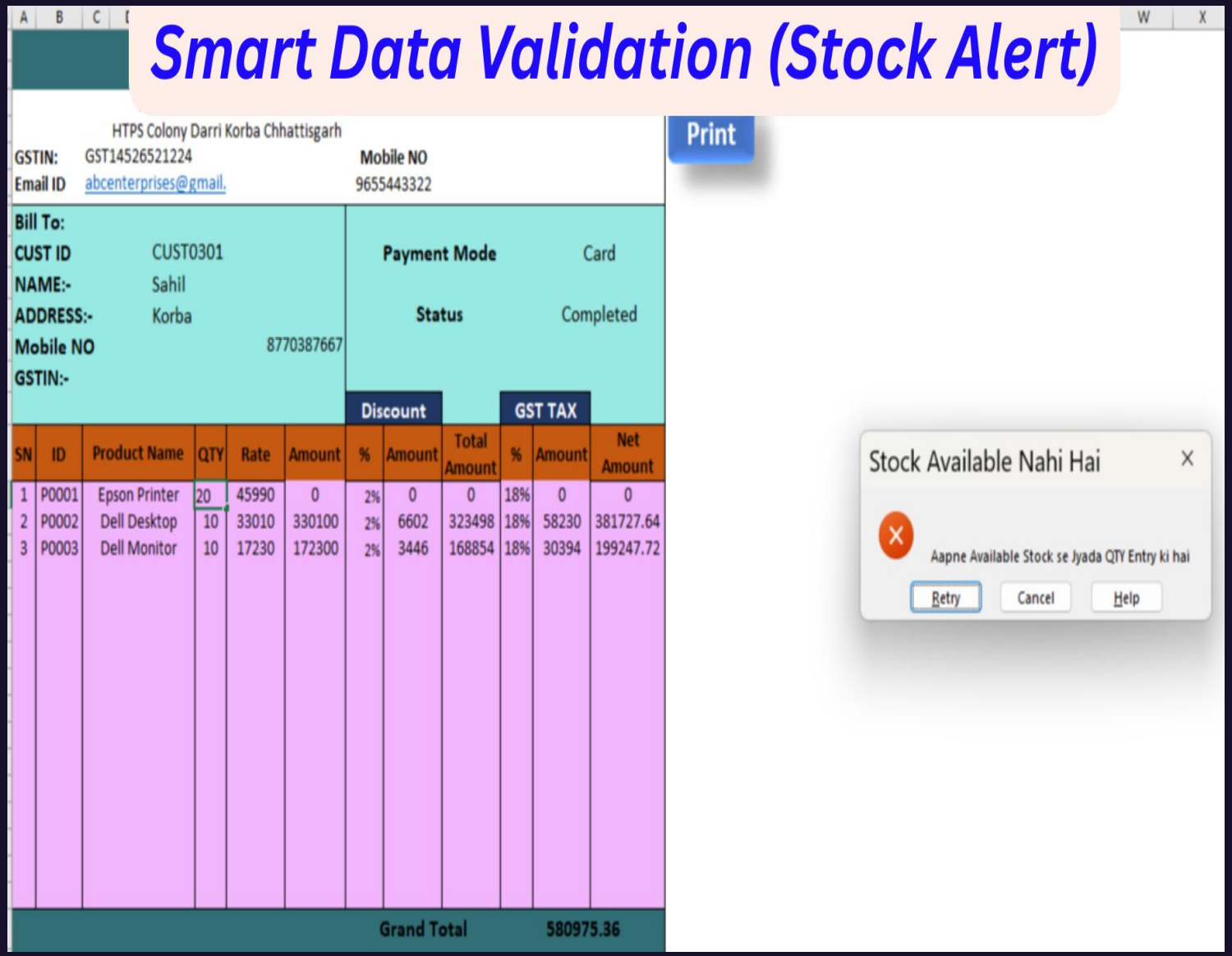
OK

"Live reorder alerts auto-fetch supplier details (SUP028 Shree Suppliers) to prevent stockouts."

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Active Safeguards Protecting Ledger and Calculation Quality

Smart Data Validation (Stock Alert)



HTPS Colony Darri Korba Chhattisgarh

GSTIN: GST14526521224 Mobile NO

Email ID abcenterprises@gmail. 9655443322

Print

Bill To:

CUST ID CUST0301 Payment Mode Card

NAME:- Sahil

ADDRESS:- Korba Status Completed

Mobile NO 8770387667

GSTIN:-

SN	ID	Product Name	QTY	Rate	Amount	Discount %	Discount Amount	Total Amount	GST TAX %	GST TAX Amount	Net Amount
1	P0001	Epson Printer	20	45990	0	2%	0	0	18%	0	0
2	P0002	Dell Desktop	10	33010	330100	2%	6602	323498	18%	58230	381727.64
3	P0003	Dell Monitor	10	17230	172300	2%	3446	168854	18%	30394	199247.72

Grand Total 580975.36

Over-Order Blocker

Blocks order processing if entered Qty exceeds Available Stock with an instant retry dialog.

ACTIVE SYSTEM CONTROL

Input Validation

Restricts numeric fields (Rates, Tax %, Quantities) against invalid text entries.

ACTIVE SYSTEM CONTROL

ID & Master Integrity

Dynamic lookup validates Customer ID and Product Code before line-item calculation.

ACTIVE SYSTEM CONTROL

Formula Protection

Critical calculation cells locked against accidental manual deletion or formula disruption.

ACTIVE SYSTEM CONTROL

High-Volume Transaction Engine Supporting 5,000+ Historical Rows

VERIFIED SYSTEM CAPABILITIES

Audit Status :

- Zero record duplication.
- Instant invoice reprint by Bill No.
- ₹0 recurring third-party software license.

1. **Baseline State**: Sales Master loaded with 5,000 historical transactions up to INV-15000 (Total: ₹405,726,484)
2. **Invoice Trigger**: Clicked "Save" macro on invoice INV-15001 for Customer Sahil.
3. **Automated Commit**: VBA instantly appended 3 line items into rows 5003,5004,5005.
4. **Instant Math Parity**: Grand Total automatically recalculated to ₹406,839,288 (+ ₹1,112,804)
5. **Audit Trail Integrity** : Permanent row entry secured with zero duplicate invoice numbers.

Before Invoice Save (Last Bill: INV-15000)

Sales Register

	INV NO.	INV Date	Customer ID	Customer Name	Product ID	Product Name	Qty	Selling Rate	Gross Amount	Disc %	Disc. Amt.	GST %	GST Amount	Total Amount	Pay Mode	Status	Net QTY	Net Amount
4984	INV-14982	31-12-25	CUST0052	Rahul Gupta	P0368	Microtek UPS	4	6450	25800	0%	0	18%	4644	30444	UPI	Cancelled	0	₹ -
4985	INV-14983	31-12-25	CUST0218	Vikas Singh	P0157	Dell Power Adapter	5	2490	12450	12%	1494	18%	1972	12928	UPI	Completed	5	₹ 12,928
4986	INV-14984	31-12-25	CUST0179	Vikas Gupta	P0137	HP Power Adapter	4	4850	19400	5%	970	18%	3317	21747	Card	Completed	4	₹ 21,747
4987	INV-14985	31-12-25	CUST0064	Ankit Kumar	P0135	Boat Speaker	8	1430	11440	12%	1373	18%	1812	11879	UPI	Completed	8	₹ 11,879
4988	INV-14986	31-12-25	CUST0111	Pooja Verma	P0139	erican TouristerLaptopE	8	840	6720	0%	0	18%	1210	7930	Cash	Completed	8	₹ 7,930
4989	INV-14987	31-12-25	CUST0152	Amit Gupta	P0316	Samsung Monitor	5	24770	123850	12%	14862	18%	19618	128606	Card	Completed	5	₹ 128,606
4990	INV-14988	31-12-25	CUST0221	Sahil Verma	P0464	Samsung Monitor	2	23660	47320	0%	0	18%	8518	55838	Online	Completed	2	₹ 55,838
4991	INV-14989	31-12-25	CUST0109	Priya Verma	P0343	Dell Laptop	2	69400	138800	10%	13880	18%	22486	147406	Online	Completed	2	₹ 147,406
4992	INV-14990	31-12-25	CUST0286	Sunita Sharma	P0009	Lg Monitor	2	21600	43200	10%	4320	18%	6998	45878	Card	Cancelled	0	₹ -
4993	INV-14991	31-12-25	CUST0285	Rahul Singh	P0168	Logitech Webcam	1	2420	2420	10%	242	18%	392	2570	Online	Completed	1	₹ 2,570
4994	INV-14992	31-12-25	CUST0189	Pooja Verma	P0368	Microtek UPS	5	6450	32250	0%	0	18%	5805	38055	Online	Completed	5	₹ 38,055
4995	INV-14993	31-12-25	CUST0180	Vikas Sharma	P0132	Kingston Ram	7	11780	82460	2%	1649	18%	14546	95357	Cash	Completed	7	₹ 95,357
4996	INV-14994	31-12-25	CUST0298	Sunita Sharma	P0432	Logitech Mouse	4	1650	6600	5%	330	18%	1129	7399	Cash	Completed	4	₹ 7,399
4997	INV-14995	31-12-25	CUST0193	Rahul Patel	P0112	HP Laptop	5	90020	450100	15%	67515	18%	68865	451450	UPI	Completed	5	₹ 451,450
4998	INV-14996	31-12-25	CUST0207	Rakesh Kumar	P0292	HP Laptop	4	88550	354200	5%	17710	18%	60568	397058	Cash	Completed	4	₹ 397,058
4999	INV-14997	31-12-25	CUST0286	Sunita Sharma	P0099	Jbl Speaker	2	7630	15260	5%	763	18%	2609	17106	Cash	Completed	2	₹ 17,106
5000	INV-14998	31-12-25	CUST0213	Sunita Gupta	P0253	Portronics USB Cable	2	200	400	0%	0	18%	72	472	Online	Completed	2	₹ 472
5001	INV-14999	31-12-25	CUST0290	Sunita Kumar	P0022	Jbl Headset	2	4610	9220	0%	0	18%	1660	10880	Online	Completed	2	₹ 10,880
5002	INV-15000	31-12-25	CUST0159	Rahul Sharma	P0067	Samsung Ssd	7	15030	105210	10%	10521	18%	17044	111733	UPI	Completed	7	₹ 111,733
5003	Grand Total																	₹405,726,484

After VBA Execution (INV-15001 Appended to Row 5003–5005)

Sales Register

	INV NO.	INV Date	Customer ID	Customer Name	Product ID	Product Name	Qty	Selling Rate	Gross Amount	Disc %	Disc. Amt.	GST %	GST Amount	Total Amount	Pay Mode	Status	Net QTY	Net Amount
4986	INV-14984	31-12-25	CUST0179	Vikas Gupta	P0137	HP Power Adapter	4	4850	19400	5%	970	18%	3317	21747	Card	Completed	4	₹ 21,747
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4988	INV-14986	31-12-25	CUST0111	Pooja Verma	P0139	erican TouristerLaptopE	8	840	6720	0%	0	18%	1210	7930	Cash	Completed	8	₹ 7,930
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4990	INV-14988	31-12-25	CUST0221	Sahil Verma	P0464	Samsung Monitor	2	23660	47320	0%	0	18%	8518	55838	Online	Completed	2	₹ 55,838
4991	INV-14989	31-12-25	CUST0109	Priya Verma	P0343	Dell Laptop	2	69400	138800	10%	13880	18%	22486	147406	Online	Completed	2	₹ 147,406
4992	INV-14990	31-12-25	CUST0286	Sunita Sharma	P0009	Lg Monitor	2	21600	43200	10%	4320	18%	6998	45878	Card	Cancelled	0	₹ -
4993	INV-14991	31-12-25	CUST0285	Rahul Singh	P0168	Logitech Webcam	1	2420	2420	10%	242	18%	392	2570	Online	Completed	1	₹ 2,570
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4996	INV-14994	31-12-25	CUST0298	Sunita Sharma	P0432	Logitech Mouse	4	1650	6600	5%	330	18%	1129	7399	Cash	Completed	4	₹ 7,399
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5002	INV-15000	31-12-25	CUST0159	Rahul Sharma	P0067	Samsung Ssd	7	15030	105210	10%	10521	18%	17044	111733	UPI	Completed	7	₹ 111,733
5003	INV-15001	06-10-26	CUST0301	Sahil	P0001	Epson Printer	10	45990	459900	2%	9198	18%	81126	531828	Card	Completed	10	₹ 531,828
5004	INV-15001	06-10-26	CUST0301	Sahil	P0002	Dell Desktop	10	33010	330100	2%	6602	18%	58230	381728	Card	Completed	10	₹ 381,728
5005	INV-15001	06-10-26	CUST0301	Sahil	P0003	Dell Monitor	10	17230	172300	2%	3446	18%	30394	199248	Card	Completed	10	₹ 199,248
5006	Grand Total																	₹406,839,288

Project Overview, Key Features & Learning Outcomes

PROJECT OVERVIEW

An Excel VBA-based Inventory & Billing ERP System designed to integrate billing , inventory management, stock validation, and sales record maintenance

KEY FUNCTIONALITIES

- Automated invoice generation with Save and Print buttons
- Automatic stock updates after invoice generation
- Low-stock alerts with supplier details
- Sales register maintenance and transaction logging
- Input validation and protection against over-ordering
- Automated amount-to-words conversion

TECHNOLOGY STACK

Microsoft Excel | VBA Macros | Data Validation | Inventory Management | Business Process Automation

KEY LEARNING OUTCOMES

- Excel VBA automation and macro development
- Multi-sheet data integration
- Inventory calculations and validation controls
- Invoice processing and sales reporting

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